

Low Power AI (LPAI) Sensors Subsystem

Application Note

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Revision history

Revision	Date	Description
AA	September 2022	Initial release

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1 Introduction

1.1 Purpose

This document describes the low power AI (LPAI) sensors subsystem and how it supports Island modes.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `cp armcc armcpp`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, `copy a:*. * b:`.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

1.3 Technical assistance

For assistance or clarification on information in this document, open a technical support case at <https://support.qualcomm.com/>.

You will need to register for a Qualcomm ID account and your company must have support enabled to access our Case system.

Other systems and support resources are listed on <https://qualcomm.com/support>.

If you need further assistance, you can send an email to qualcomm.support@qti.qualcomm.com.

2 Island concept

Island or Low Power Island mode supports always-on low power use cases. LPAI DSP on SM8550 (and future targets) supports Island where the DSP processor works self sufficiently and does not access external memory, such as DDR, but uses internal memories, like TCM and LLCC. Power consumption is reduced when DDR goes into self-refresh mode and other internal buses on SoC required to access non-island modules are turned off. Low power modes, such as XO shutdown and power collapse, are supported in Island.

In Island mode, mode LPAI DSP runs in a power optimized way and hence at reduced speeds.

In Normal mode, LPAI DSP has access to all external memories, like DDR and other chip resources. This mode is used when interactions outside of DSP are required. Here, performance-intensive operations are possible; DSP can run at full speed and vote for on-chip resources.

Island code is a subset of the entire image and can run in Island mode or Normal mode.

During Island mode of operation, code and data needed for operation are copied from DDR to L2/TCM/LLCC.

LPAI DSP provides:

- Hardware support to isolate Low Power Island with separate LPI_MX and LPI_CX power rails
- Build system automation to generate Island image:
 - SCons builder support to specify Island contents
 - Individual functional units specify their Island contents in SCons script
- XML configuration to define Island size and other Island parameters
- Autogeneration of Linker script with appropriate “Island” sections based on SCons script definitions
- Image builder tool to postprocess the image and prepare Island segments
- QuRT™ APIs to provide OS support
- Snapdragon™ Sensors Core (SSC) solution, which supports its own way to easily configure something in Island or Normal mode. Refer to respective target *Sensors Overview* document for more details.

The sensor subsystem, which is part of LPAI Island, supports autonomous highly power-efficient operation of sensor interfacing and processing that meets aggressive low power goals of today's market requirement. It has the following hardware modules:

- QUPv3 consisting of serial interface for SPI, I²C/I³C, and UARTs with GSI
 - SPI Slave mode is not supported in SSC
- Local sensor SRAM (PRAM) for sensor data storage in Island mode
- MCC block consisting of clock controller, IO mode mux, and software registers
- Context change detector (CCD) for vector processing – Low power subsystem for detecting the change in device states. Example to detect when the device transitions from absolute rest to motion.
- IBI controller – Enables detection and processing of IBI while the SSC is in power collapse

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3 Build support for Island configuration

The Island specification defines what is supported within Island. This consists of:

- Island code/data
- DSP interrupts required by Island
- HWIO memory ranges required by Island

The Island specification can be defined statically and currently this is supported through the LPAI DSP build infrastructure:

- SCons
 - Provision to include the files/libraries (or sections within them) into ulmage
- LCS
 - Defines the placement of various ulmage sections in the full image
- XML
 - Defines the Island pool with the TCM and DDR partitions
 - Determines the sections that have to be picked by each of the Island partitions

Examples

- SCons

File: `adsp_proc\sscf\framework\build\framework.scons`

```
if 'USES_MEMSEG' in env or 'USES_ISLAND' in env:
    sections = ['.text.sns', '.data.sns', '.rodata.sns',
               '.rodata..lswitch.table.*']
    if 'USES_MEMSEG' in env:
        env.TagLibraryForMemorySegment(env['SSC_BUILD_TAGS'], 'ssc', lib, sections)
        env.TagLibraryForMemorySegment(env['SSC_BUILD_TAGS'], 'ssc', my_static_lib,
        sections)
        if len(VARIANT_PROTO_SRCFILES) > 0:
            env.TagLibraryForMemorySegment(env['SSC_BUILD_TAGS'], 'ssc', proto_lib)
    elif 'USES_ISLAND' in env:
        env.AddIslandLibrary(env['SSC_BUILD_TAGS'], lib, sections)
        env.AddIslandLibrary(env['SSC_BUILD_TAGS'], my_static_lib, sections)
        if len(VARIANT_PROTO_SRCFILES) > 0:
            env.AddIslandLibrary(env['SSC_BUILD_TAGS'], proto_lib)
```

■ LCS (Linker)

```
/* QURT_ISLAND_UKERNEL */
. = ALIGN(4K);
.data.common.island : {
    /* COMMON_DATA */
    /* QURTOS_DATA */
}
. = ALIGN(4K);
.data.ssc.island : {
    /* SSC_DATA */
}
. = ALIGN(4K);
.data.va.island : {
    /* VA_DATA */
}
. = ALIGN(4K);
```

■ XML config

```
<build>
    <section name="*.common.island" physpool="TCM_PHYSPOOL"/>
    <section name="*.ssc.island"
physpool="SSC_ISLAND_LPASS_TCM_POOL"/>
    <section name="*.va.island"
physpool="TCM_PHYSPOOL,AUDIO_ISLAND_LPASS_TCM_POOL"/>
    <section name="*.qshtech.island"
physpool="SSC_ISLAND_LPASS_TCM_POOL"/>
    <section name="*.camera.island"
physpool="CAM_LLCC_ISLAND4_POOL,CAM_LLCC_ISLAND1_POOL"/>
    <section name="*.qsh.island"
physpool="SSC_ISLAND_LPASS_TCM_POOL"/>
</build>
```


■ Island pool config

```

    <physical_pool name="CAM_LLCC_ISLAND1_POOL" island="true"
collapse="tcm">
    <region base="0xCFFA40000" size="0x1C0000"
cache_policy="l1_wb_l2_cacheable"/>        <!--SCID 12 768kB -->
    </physical_pool>

    <physical_pool name="CAM_LLCC_ISLAND4_POOL" island="true" collapse="tcm">
    <region base="0xFFFFC00000" size="0x400000"
cache_policy="l1_wb_l2_cacheable"/>        <!--SCID 15 - 256kB -->
    </physical_pool>

    <physical_pool name="TCM_PHYSPOOL" island="true" collapse="tcm">
<!-- 1.5MB common pool -->
    <region base="0x06400000" size="0x180000"
name="QURTOS_ISLAND_REGION_TCM" cache_policy="writethrough"/>
    </physical_pool>
    <physical_pool name="AUDIO_ISLAND_LPASS_TCM_POOL" island="true"
collapse="tcm">        <!-- 896KB Audio only pool -->
    <region base="0x06080000" size="0xE0000"
cache_policy="l1_wb_l2_cacheable"/>
    </physical_pool>
    <physical_pool name="CAM_ISLAND_LPASS_TCM_POOL" island="true"
collapse="tcm">        <!-- 0.25MB Camera only pool -->
    <region base="0x06160000" size="0x40000"
cache_policy="l1_wb_l2_cacheable"/>
    </physical_pool>
    <physical_pool name="SSC_ISLAND_LPASS_TCM_POOL" island="true"
collapse="tcm">        <!-- 1.870MB SSC only pool -->
    <region base="0x061A0000" size="0x1DF000"
cache_policy="l1_wb_l2_cacheable"/>
    </physical_pool>
    <physical_pool name="SSC_SDC_ISLAND_LPASS_TCM_POOL" island="true"
collapse="tcm">        <!-- 4KB SSC only pool -->
    <region base="0x0637F000" size="0x1000" cache_policy="uncached"/>
    </physical_pool>

```

■ Type of Island specification supported

File: core/api/power/uSleep/islands/default_multi/adsp/uSleep_islands.h

```

#define USLEEP_ISLAND_LLC          ISLAND_SPEC(0)
#define USLEEP_ISLAND_STD          ISLAND_SPEC(1)

```

Sample Island specification:

adsp_proc\config\chipset\kailua\adsp\island_spec_core.xml

```

<irq num="0"/> <!-- PMI -->
<irq num="2"/> <!-- q6ss_qtmr_irq[0] -->
<irq num="4"/> <!-- qTimer frame 3 -->
<irq num="31"/> <!-- CTI interrupt -->
<irq num="32"/> <!-- HW DOG Bark interrupt -->
<irq num="36"/> <!-- bi_mx_lpass_4_aoss_mx_gpio_172 -->
<irq num="40"/> <!-- bi_mx_lpass_11_aoss_mx_gpio_179 -->
<irq num="41"/> <!-- lpass_rsc_core_comp_irq -->
<irq num="44"/> <!-- bi_mx_lpass_1_aoss_mx_gpio_169 -->
<irq num="45"/> <!-- bi_mx_lpass_12_aoss_mx_gpio_180 -->
<irq num="312"/>
<memory_phys base="0x06880000" size="0x40000"/> <!-- Covers QTMR register
-->
<memory_phys base="0x25AA0000" size="0x10000"/> <!-- Covers LLC region -->
<memory_virt base="0xe0000000" size="0xF0000"/> <!-- CLK_CNTL -->
<memory_virt base="0xee000000" size="0xC00000"/> <!-- LPASS -->
<memory_phys base="0x07220000" size="0x10000"/> <!-- LPASS_ICB_REGION for
Island-to-island-->
<memory_phys base="0x07100000" size="0x100000"/> <!-- LPASS_STM stimulus
ports region -->
<memory_phys base="0x07250000" size="0x10000 "/> <!--
LPASS_MCC_LPASS_REG -->
<memory_virt base="0xef800000" size="0x400000"/> <!-- SSC CLK_CNTL -->
<memory_virt base="0xee000000" size="0x100000"/> <!-- LPASS CLK_CNTL -->
<physical_pool name="TCM_PHYSPOOL"/>
<dynamic_irqs num="6"/> <!-- GPIO interrupts -->
<dynamic_memranges num="5"/> <!-- QDSS HWIO regions -->

```

A References

A.1 Related documents

Title	Number
Qualcomm Technologies, Inc.	
<i>SM8550 Sensors Overview</i>	80-35348-24

A.2 Acronyms and terms

Acronym or term	Definition
CCD	Context change detector
LPAI	Low power AI
SSC	Snapdragon Sensors Core